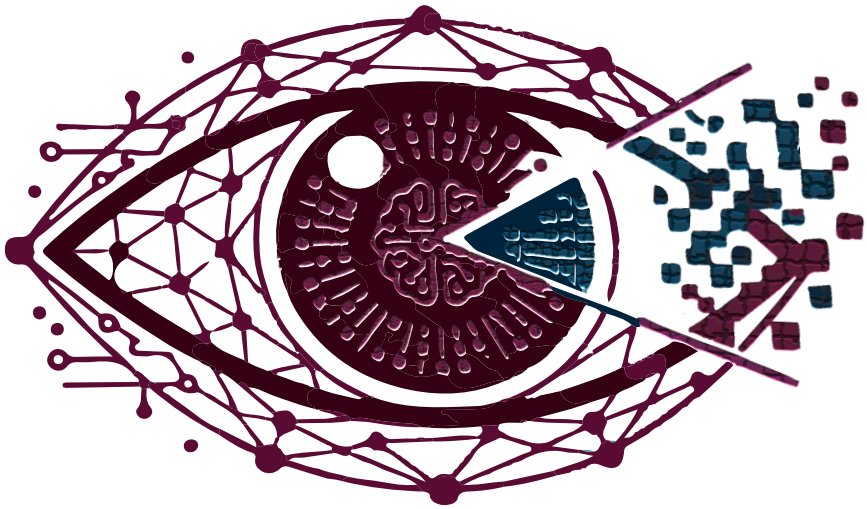
TRUTHLENS

Deepfake Detection Platform



Феникс...

Project Overview:

The ***TruthLens*** project aims to develop a system that uses artificial intelligence and computer vision techniques to detect AI-generated images and distinguish them from real images. The importance of this project stems from the rapid advancement of generative AI models, which have become capable of producing highly realistic images that can sometimes be difficult to distinguish from authentic images through visual inspection alone.

The current version of the project focuses on developing an image classification model, along with a user interface that allows users to upload an image, analyze it, and view the predicted result. The system is intended to serve as an assistive tool for verifying the authenticity of visual content, while evaluating its performance and assessing the limits of its reliability through practical experiments.

Problem Statement:

The rapid advancement of generative artificial intelligence technologies has made it easier to produce highly realistic images, increasing the difficulty of distinguishing real images from artificially generated ones. This development poses challenges for verifying the authenticity of digital content, particularly when images are shared across online platforms and various information sources.

The main challenges associated with this phenomenon include:

- Media Misinformation: AI-generated images may be used to spread misleading information or support false narratives.
- Identity Impersonation and Digital Abuse: Synthetic images may be created and used to harm individuals or damage their reputations.

- Digital Fraud: AI-generated images may be exploited in fraudulent activities that rely on visual content.
- Detection and Generalization Challenges: The effectiveness of some detection systems may decline when they are tested on images generated using models or techniques that were not represented in the training data.

Accordingly, the research problem lies in developing and evaluating a system capable of distinguishing between real images and AI-generated images, while investigating its ability to handle variations across image sources and identifying the limitations that may affect its performance.

The scope of the current version of ***TruthLens*** is limited to classifying images as either real or AI-generated. The system is not intended to detect all forms of localized image manipulation or every type of deepfake.

Project Objectives:

The ***TruthLens*** project aims to achieve the following objectives:

1. Develop an Image Classification Model: Build an artificial intelligence model capable of distinguishing between real images and AI-generated images.
2. Prepare Suitable Training Data: Inspect, clean, preprocess, and split the dataset using a methodology that minimizes preprocessing errors and data leakage.
3. Scientifically Evaluate Performance: Measure the model's performance using appropriate classification metrics, and analyze errors and limitations observed during experiments.
4. Investigate Generalization Capability: Assess the extent to which the model maintains its performance when processing data that differs from the training data, within the limits of the available tests.

5. Develop a Clear User Interface: Create an interface using *Streamlit* that allows users to upload images and view the results in a simple and understandable manner.
6. Clarification of Output Limitations: Present classification results and available model confidence scores without treating them as definitive proof of an image's authenticity or artificial origin.

Project Significance:

The significance of ***TruthLens*** lies in the following scientific and practical aspects:

I. Scientific Significance in Computer Vision:

Detecting AI-generated images represents one of the important challenges in computer vision. Synthetic images may contain visual patterns or digital artifacts that can be used to distinguish real content from generated content. However, these indicators may vary depending on the generative models, image quality, and processing methods, making the generalization of detection systems an ongoing research challenge.

This project contributes to the study of this problem by developing an image classification model and evaluating its performance using appropriate scientific metrics, helping to improve the understanding of the capabilities and limitations of detection techniques.

II. Practical and Ethical Significance:

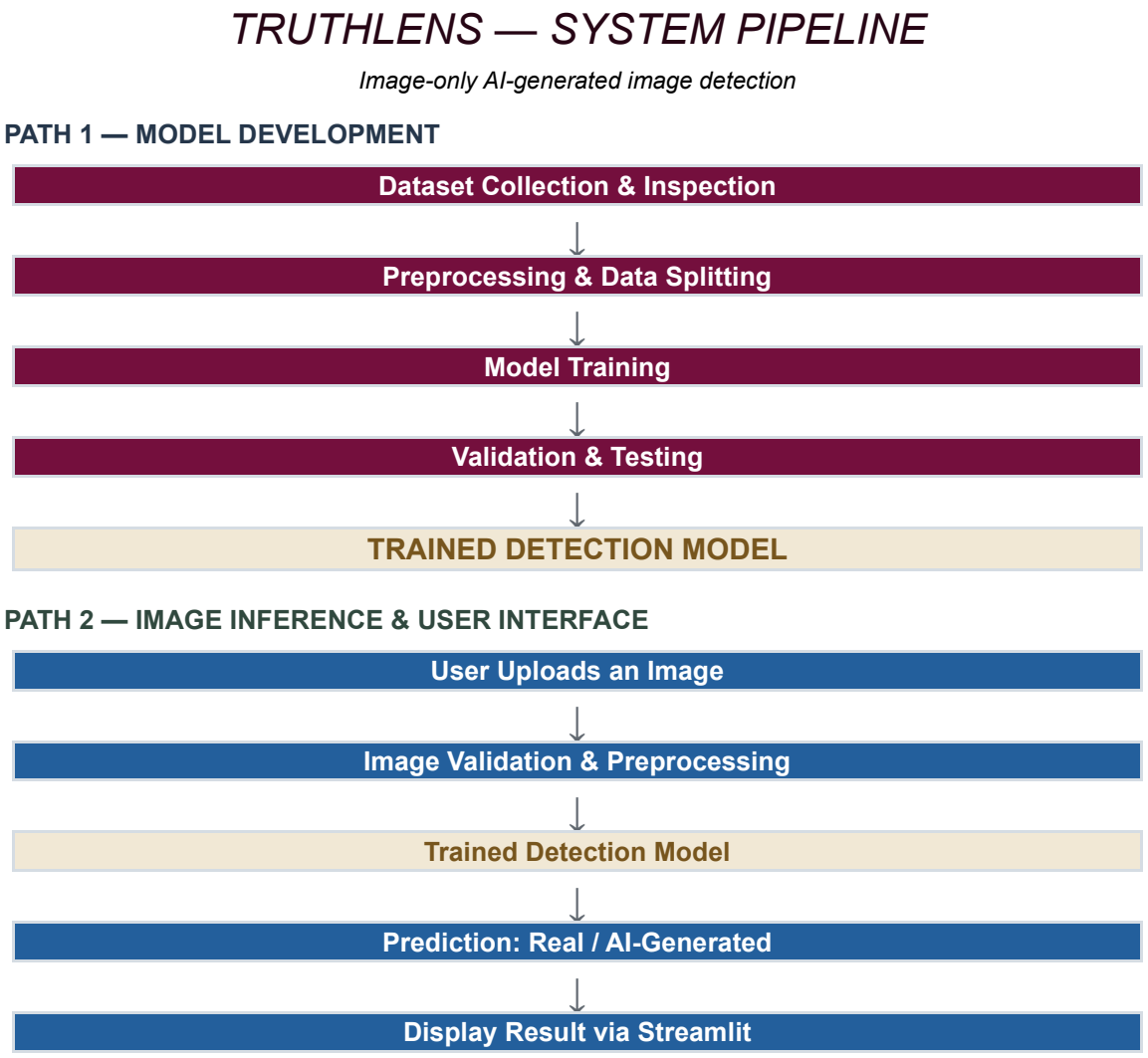
The system can provide users with an assistive tool for examining digital images and promote awareness of the importance of verifying content before relying on it or sharing it. Nevertheless, the system's results should not be considered definitive evidence of an image's authenticity or artificial origin. Instead, they should be interpreted in light of the model's performance and the limitations of the training and testing data.

Literature Review:

Paper Title & Authors	**Publisher**	**Technique**	**Datasets Used**	**Performance Metrics**
Detecting Deepfakes with Self-Blended Images (Japan...2022)	*CVPR (IEEE/CVF*	• Self-Blended Images (SBIs) synthetic data generation blending pseudo source/target from single real image. • Backbone classifiers: EfficientNet-B4, ResNet-34, Xception.	Training: Pristine real images from FF++. Testing: FF++, Celeb-DF-v2, DFD, DFDC, DFDCP, FFIW-10K.	Video/Frame AUC / Ap: • FF++ AUC: 99.64% • Celeb-DF AUC: 93.18% • DFD AUC:97.56% • DFDC AUC:72.42% • DFDCP AUC: 86.15%
Transcending Forgery Specificity with Latent Space Augmentation for Generalizable Deepfake Detection (China / United States…2024)	CVPR (IEEE/CVF)	• Latent Space Data Augmentation (LSDA). • Within-Domain (WD) noise perturbations & Cross-Domain (CD) mixup in latent space. • Teacher-Student distillation with EfficientNet-B4 backbone.	Training: FF++ c23. Testing: Celeb-DF-v1/v2, DFDCP, DFDC, AIGC GAN/Diffusion datasets.	AUC / AP / EER: • Avg DeepfakeBench AUC: 82.6% • CDF-v1 AUC: 86.7% • CDF-v2 AUC: 83.0% (Video AUC: 91.1%) • DFDC AUC: 73.6% (Video AUC: 77.0%)
DF40: Toward Next-Generation Deepfake Detection (China...*2024*)	NeurIPS (Datasets & Benchmarks)	• DF40 Benchmark & Dataset: 40 forgery techniques across Face Swapping, Reenactment, Entire Synthesis, and Editing. • Evaluated 8 SoTA detectors across 4 standard protocols	Benchmark Dataset: DF40 (100K+ videos, 1M+ images including DeepFaceLab, HeyGen, MidJourney-6, DiT, PixArt-α, etc.)	Frame/Video AUC: • Traditional detectors suffer severe drops on open-set forgeries (Xception avg AUC: 54.6%, SBI: 51.4%, RFM: 64.8%). • CLIP baseline achieved avg open-set AUC: 69.2%
FakeRadar: Probing Forgery Outliers to Detect Unknown Deepfake Videos (China…*2025*)	ICCV (IEEE/CVF)	• Forgery Outlier Probing (FOP): Dynamic GMM subcluster modeling & cluster-conditional outlier generation[1]. • Outlier-Guided Tri-Training (OGTT): Triplet-class classification ("Real", "Fake", "Outlier")[5][51]. • CLIP ViT-B/16 backbone with ST-Adapter	Training: FF++ HQ. Testing: Celeb-DF-v2, DFDCP, DFDC, DFD	Video-level AUC: • Cross-dataset Avg AUC: 90.1% • CDFv2 AUC: 91.7% • DFDC AUC: 84.1% • DFDCP AUC: 88.5% • DFD AUC: 96.2%
Community Forensics: Using Thousands of Generators to Train Fake Image Detectors (United States University of Michigan…2025)	CVPR 2025	Trains image-forensics classifiers using a large and diverse collection of generated images from thousands of generative models, including CNN-based and Vision Transformer (ViT) approaches.	Community Forensics (2.7M generated images from 4,803 models); Wang et al.; Ojha et al.; Synthbuster; GenImage	The proposed model achieved 0.977 mean mAP and 0.966 mean accuracy across the evaluated benchmarks. On the high-resolution setting, it achieved 0.994 mean mAP and 0.986 mean accuracy.
PPM-CLIP: Probabilistic Prompt Modeling for Generalizable AI-Generated Image Detection (2026)	CVPR (IEEE/CVF)	• CLIP-based vision-language model. • Probabilistic Prompt Modeling (PPM) for adaptive prompts. • Patch-Wise Contrastive Learning (PWCL) to enhance sensitivity to subtle image artifacts	Ojha, GenImage, DRCT	Accuracy: • Ojha: 98.8% • GenImage: 99.6% • DRCT: 95.72%

Technical Methodology:

The proposed methodology for the ***TruthLens*** project consists of a series of interconnected stages, beginning with data preparation and ending with presenting the classification result to the user.



I. Dataset Collection and Inspection:

Use a suitable dataset containing real images and AI-generated images. Examine its structure, labels, and characteristics, and verify that the samples are suitable for training and evaluation.

II. Preprocessing:

Load the images and verify their validity, then standardize their dimensions and representation to meet the requirements of the selected model. Care is taken to preserve important visual information and apply preprocessing operations appropriate to the nature of the data.

III. Dataset Splitting:

Divide the dataset into training, validation, and testing sets using an appropriate strategy that minimizes information leakage between the sets and helps assess the model's generalization capability.

IV. Model Training:

Select a suitable model for the classification task and train it on the training data. Use the validation set to monitor performance and adjust the model's settings when necessary.

V. Inference and Classification:

Accept a new image and preprocess it using the same procedure applied during training. Then pass it to the model to obtain a prediction for one of the two classes: real image or AI-generated image.

VI. Model Evaluation:

Evaluate the model on data that was not used during training and measure its performance using appropriate metrics, such as *Accuracy*, *Precision*, *Recall*, and *F1-score*, in addition to *ROC-AUC* when appropriate for the evaluation method.

VII. User Interface:

Develop a user interface using ***Streamlit*** that allows users to upload an image and view the classification result, along with relevant information about the model's outputs and limitations.

These stages represent an initial methodology that can be further refined after inspecting the dataset and determining the final model and experimental settings.

Dataset — Preliminary Description:

The *TruthLens* project provisionally uses the *CommunityForensics-Small* dataset, published on the *Hugging Face* platform by *OwensLab*. This dataset was selected to investigate the task of distinguishing between real images and AI-generated images.

Before finalizing the dataset description, its actual structure and characteristics will be examined, including the number of available samples, class distribution, image and label storage formats, and access and storage requirements. The preprocessing and data-splitting strategies will be determined based on the findings of this inspection, with any subset used in the experiments properly documented.

Methodological Note: Figures and performance metrics reported in previous studies are not results of the *TruthLens* project. The project's results will be presented after conducting the experiments and evaluating the proposed model.

Dataset Description:

The **CommunityForensics-Small** dataset, available on the Hugging Face platform, is designed for studying real images and AI-generated images. It features diverse content and sources, including human faces, natural scenes,

objects, and various other subjects, as well as images generated using a wide range of image-generation models.

Dataset Size: According to the dataset description, it contains approximately 556,000 images, evenly distributed between 278,000 real images and 278,000 AI-generated images.

Image Dimensions and Color Mode: The image dimensions specified in the dataset description are 512 × 512 pixels. The images use the RGB color mode, which consists of three colored channels: red, green, and blue.

Image Classification: The dataset contains two main classes: real images and AI-generated images. Each image is associated with a label identifying its corresponding class.

Dataset Splitting and Balance: The data is provided in a single split named train, which means it is not pre-divided into separate training, validation, and testing sets. According to the reported figures, the dataset is balanced between the two classes, with equal numbers of real and AI-generated images. Therefore, no class imbalance is indicated in the reported distribution.

Source: OwensLab, *CommunityForensics-Small*, Hugging Face Datasets: https://huggingface.co/datasets/OwensLab/CommunityForensics-Small

Contents

Project Overview: 1

Problem Statement: 1

- Media Misinformation: 1
- Identity Impersonation and Digital Abuse: 1
- Digital Fraud: 2
- Detection and Generalization Challenges: 2

Project Objectives: 2

Project Significance: 3

I. Scientific Significance in Computer Vision: 3

II. Practical and Ethical Significance: 3

Literature Review: 4

Technical Methodology: 5

I. Dataset Collection and Inspection: 6

II. Preprocessing: 6

III. Dataset Splitting: 6

IV. Model Training: 6

V. Inference and Classification: 6

VI. Model Evaluation: 6

VII. User Interface: 7

Dataset — Preliminary Description: 7

Dataset Description: 7